Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Cross-Border Analytics, Business Intelligence Infrastructure, Dimensional Data Modeling, A/B Testing & Experimentation, Predictive Modeling & Forecasting, Automated Data Quality Audits & Firewalls, Machine Learning Anomaly Detection, Generative AI & Prompt Engineering, Data Governance & Semantic Layers, KPI Instrumentation.

Programming Languages & Tools: AWS (S3, Redshift), Azure Cloud (ADF, Synapse, Azure SQL, Databricks), Snowflake, Apache Spark / PySpark, Python (Scikit-Learn, XGBoost, SARIMA), Power BI (DAX), Tableau, MicroStrategy, KIRO IDE, SQL.

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Business Intelligence & Analytics Engineering Manager, Cross Border Science — **May 2022 – Present**

- **Led and mentored high-performing teams** delivering production-grade BI pipelines, dimensional data modeling (Star/Snowflake schemas), and self-service dashboards powering cross-border science initiatives.
- **Architected high-throughput data processing workflows** and automated data quality audit frameworks with firewalls on AWS (S3, Redshift) and Azure, reducing operational turnaround by over 80%.
- **Engineered advanced predictive and forecasting models** (SARIMA, XGBoost, Random Forests) to optimize workforce capacity planning across marketplace geographies, improving operational reliability and response times by 20%.
- **Spearheaded SpectrumIQ conversational AI and automated QA frameworks** leveraging Agentic AI and prompt engineering in KIRO IDE to cut manual testing turnaround by 50% and accelerate delivery timelines by 40%.
- **Established proactive ML-based anomaly detection systems** with 95% accuracy to monitor data freshness, drift, and pipeline health, maintaining 99.9% data availability SLA across critical executive KOM dashboards.

Master of Science in Information Systems & Data Science Graduate — **May 2022 – Present**

- **Completed advanced academic coursework** in information systems and data science at University of Cincinnati, graduating with a cumulative GPA of 3.95/4.0.
- **Developed foundational expertise in statistical modeling** machine learning algorithms, and database management systems to solve complex analytical challenges.
- **Applied advanced data analytics techniques** to derive actionable business intelligence from large-scale structured and unstructured datasets.

Bachelor of Engineering in Petrochemical — **May 2022 – Present**

- **Graduated with a Bachelor of Engineering** from Pune University MIT, establishing a strong analytical foundation in mathematics, systems engineering, and problem-solving.
- **Applied rigorous quantitative methodologies** to optimize complex industrial processes and interpret technical performance metrics.
- **Demonstrated technical versatility** transitioning core analytical and engineering problem-solving principles into high-impact data science and BI domains.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.

- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.